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Predicting IMO Performance from Prior International Olympiad Results

Phase 1: dataset, descriptive statistics and baselines

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Motivation and Research Questions

- The **IMO**: ∼600 pre-university contestants a year, 6 proof problems × 7 points, about half receive medals
- Results are used as a **measuring instrument**: to select national teams and, recently, to benchmark LLM reasoning
- Both uses assume a result reflects a **stable ability**, not one noisy exam
- Most contestants compete **before** the IMO: JBMO, BMO, EGMO, RMM, APMO, MEMO, PAMO
- **Gap**: prior work is descriptive (Berg, 2021), country-level, or predicts careers *from* the IMO (Agarwal & Gaulé, 2020); no study predicts individual IMO results from other olympiads

Research questions

RQ1 How accurately can a contestant's IMO **score and medal** be predicted from their prior international results and their country's history?

RQ2 Which prior competitions carry the most **predictive signal**?

RQ3 For whom do predictions **fail**, and are the errors systematic?

Using **only information available before** each IMO.

Data Acquisition: Sources and Crawling

Source	Content	How it is collected	Requests
`imo-official.org`	IMO 2000–2026: problem scores, rank, award, official contestant id	one page per year; JSON embedded in the HTML	27
`moresults.org`	9 olympiads, 1959–2026; profiles linking a person's participations	client-side app: data file found in its JS bundle, downloaded once (not 21,590 pages)	4
JBMO organisers	JBMO 2018–2020, 2022 (missing above); 2020 with every contestant	PDFs + HTML; codes (`ROM2`) mapped to the team pages	40

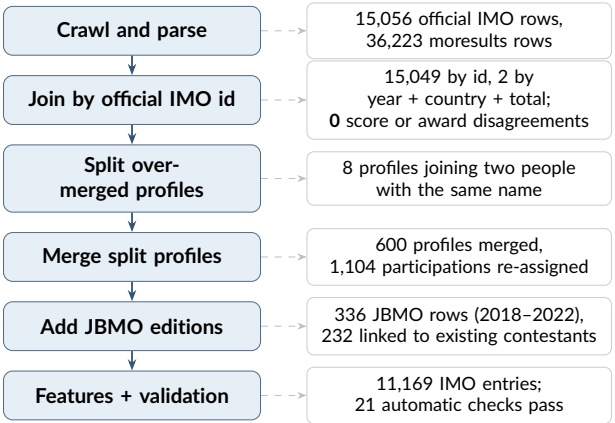
Crawling etiquette

- `robots.txt` checked: both sites allow crawling
- Descriptive user agent, 3 s between requests
- Local cache: no page is requested twice

Privacy

- Names used only for linking, inside a local cache
- Published tables: salted-hash ids, no names
- Source PDFs kept locally, cited, not committed

Building the Dataset: Record Linkage



Split profiles: moresults.org creates two profiles when a name is written differently:

- *Gualdi Pietro* (RMM) = *Pietro Gualdi* (BMO, IMO)
- *Avădănei* = *Avadanei*

Merged if: same name up to order and accents, same country, never in the same edition, career $\leq$ 7 years. **14/14** sampled merges checked by hand were correct.

Over-merged profiles: the opposite error, e.g. one *Takuya Inoue* at IMO 2007 and another at IMO 2014–2016. Profiles spanning > 7 years are split at gaps of > 3 years.

Neutral team: Russia competed without a flag in 2022–2025 (24 rows) $\rightarrow$ assigned `RUS`.

Dataset Design

Raw tables: relational

Table	One row per	Rows
`participations`	person × edition	36,564
`competitions`	edition	178
`contestants`	person	21,105
`countries`	country	157

Final table: one row per contestant per IMO

IMO editions	2008–2026
Rows	11,169
Contestants / countries	7,147 / 133
`full`: columns (features)	114 (97)
compact: columns (features)	29 (20)

Every column defined in `docs/data_dictionary.md`

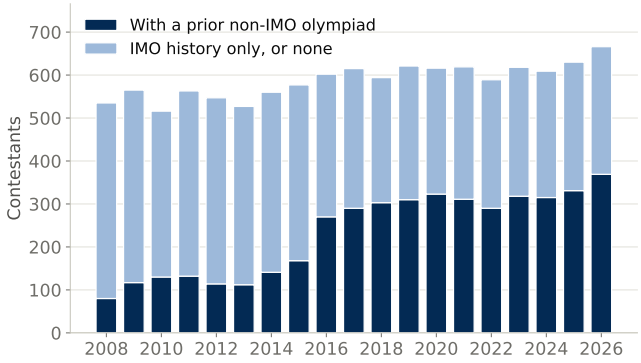
Columns grouped by prefix

`target_`	total (0–42), award, percentile, z-score, P1–P6
`imo_`	earlier IMOs: attempt number, best and last result
`hist_`	all earlier events: count, best award, mean percentile, trend
`jbmo_` … `pamo_`	one block per olympiad: status, best score, percentile, award
`team_`	IMO team that year: size, rank in team by history
`country_`	previous 3 IMOs: mean score, medal rate, regional access
`context_`	the IMO: host country, online (2020–21)
`id_`, `meta_`	not features: ids; gender proxy, continent

No leakage, explicit missing values

Only events **before** the IMO: earlier years, or the same year at RMM, APMO, EGMO, BMO, JBMO. No history → `NaN`, never 0; `<c>_status` says why: not held, no data, or did not take part.

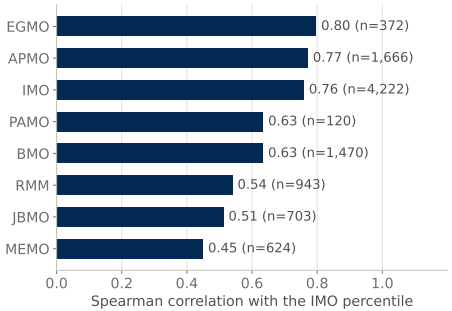
Descriptive Statistics: Who Has a History?

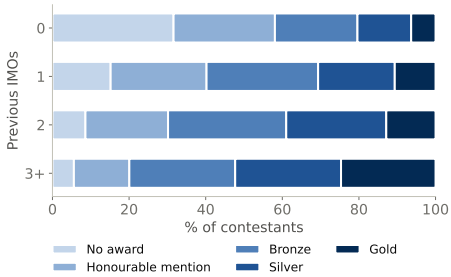


IMO contestants per year, by prior international history.

- **58.6%** of IMO entries have an earlier international result; **39.6%** an earlier olympiad other than the IMO
- That share grows from **15%** (2008) to **55%** (2026), as regional olympiads and their data appear
- Most common prior events: an earlier **IMO** (37.8%), **APMO** (14.9%), **BMO** (13.2%)
- Awards are balanced except gold: none 24.5%, HM 25.3%, bronze 24.5%, silver 17.1%, gold 8.6% $\rightarrow$ macro-averaged metrics

Descriptive Statistics: Does Earlier Performance Carry Over?





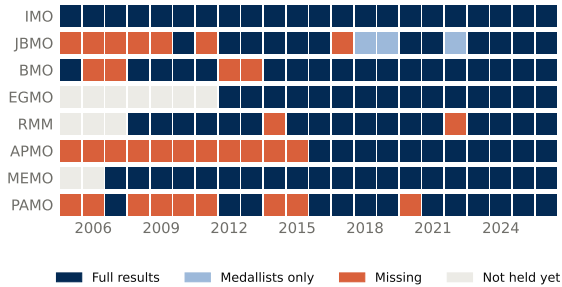
Best earlier percentile at each competition vs IMO percentile:

- **EGMO** (0.80), **APMO** (0.77) track the IMO as closely as an earlier IMO (0.76)
- Junior and regional contests less: JBMO 0.51, MEMO 0.45

IMO award by number of earlier IMOs:

- Gold: **6%** at a first IMO, **25%** at a fourth or later
- Part of the signal is the country: strong countries dominate APMO and EGMO

Data Quality: Validation and Coverage



21 automatic checks, each recomputed independently:

- one row per contestant and edition; scores within range
- **no future event** in any feature
- country features and targets equal to a direct recomputation
- every column documented

Known limitations:

- Missing editions → `status = no_data`; APMO lists only the top 10 per country
- Month-level dates: same-year events use the usual month
- **Selection**: only IMO contestants are observed

Fixed during the data audit

- 600 split and 8 over-merged profiles
- country modal award computed on 1 year instead of 3
- unstable trend from events weeks apart

Baseline Models

Temporal split: train on IMO 2008–2022 (8,646), test on IMO 2023–2026 (2,523); 95% bootstrap intervals over test contestants. Regressions use the 20 compact features.

Score (0–42)

Model	MAE	R^2
Training mean	8.37	-0.04
Country, last 3 IMOs	4.97	0.60
Linear regression	**4.36** [4.23, 4.49]	**0.70**

Award (5 classes)

Model	Accuracy	$\pm$**1 award**	Macro-F1
Majority class	0.25	0.74	0.08
Country's modal award	0.47	0.91	0.46
Logistic regression	**0.52**	**0.91**	**0.51**

- The **country** alone gives $R^2 = 0.60$; individual history adds **+0.10** (top coefficients: country mean, IMO attempt, rank in team)
- Exact medal for 52% (macro-F1 0.51 [0.49, 0.53]), at most one step off for 91%; silver is the hardest class
- Contestants **without** history are predicted better (MAE 4.17 vs 4.46): their scores are lower and less spread

Intended Analyses (Phase 2)

	Planned analysis
RQ1	Random forest, gradient boosting (LightGBM) on all 97 features; an ordinal model for the award; same temporal split and bootstrap intervals
RQ2	SHAP values aggregated per feature block (`jbmo_`, `bmo_`, …); ablation removing one competition at a time
RQ3	Residuals by continent, regional access, prior history, `status = no_data`, gender proxy; the largest over- and under-predictions

Also planned

- Predicting z-scores, which remove year-to-year difficulty
- Per-problem targets: is the hard P3 / P6 predictable?
- Repeat contestants: how much does one IMO predict the next?
- Report (max. 10 pages) and final presentation

Code and data: `github.com/AnaBogdan05/imo-results-predictor`

Thank you for your attention!

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